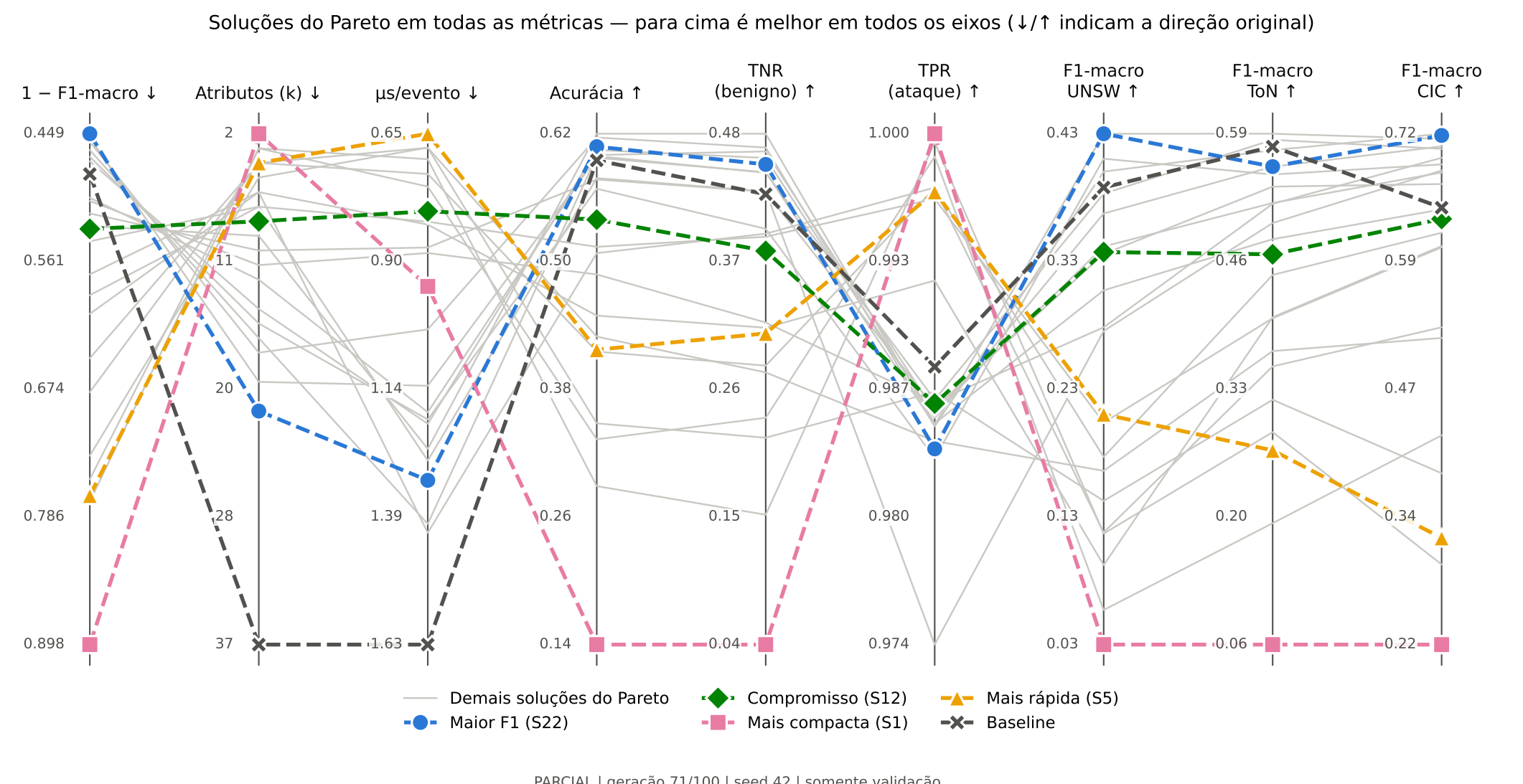
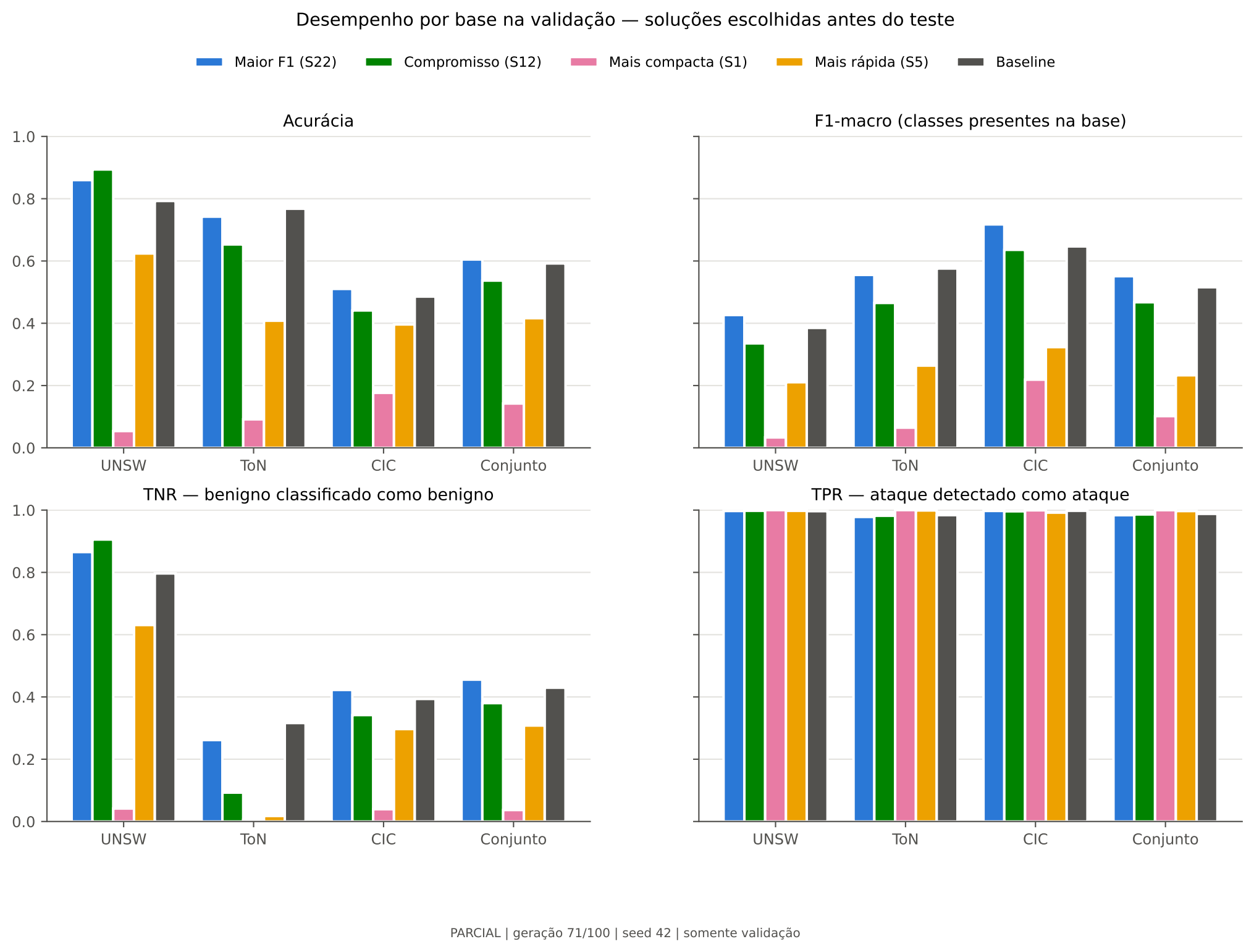
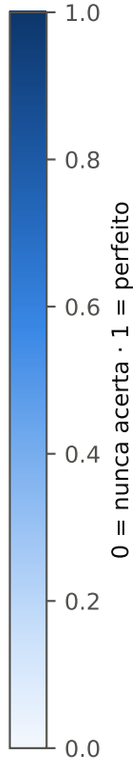

Soluções do Pareto em todas as métricas — para cima é melhor em todos os eixos (↓/↑ indicam a direção original)
1 − F1-macro ↓
Atributos (k) ↓
µs/evento ↓
Acurácia ↑
TNR (benigno) ↑
TPR (ataque) ↑
F1-macro UNSW ↑
F1-macro ToN ↑
F1-macro CIC ↑
0.449
0.561
0.674
0.786
0.898
2
11
20
28
37
0.65
0.90
1.14
1.39
1.63
0.62
0.50
0.38
0.26
0.14
0.48
0.37
0.26
0.15
0.04
1.000
0.993
0.987
0.980
0.974
0.43
0.33
0.23
0.13
0.03
0.59
0.46
0.33
0.20
0.06
0.72
0.59
0.47
0.34
0.22
Demais soluções do Pareto
Compromisso (S12)
Mais rápida (S5)
Maior F1 (S22)
Mais compacta (S1)
Baseline
PARCIAL | geração 71/100 | seed 42 | somente validação

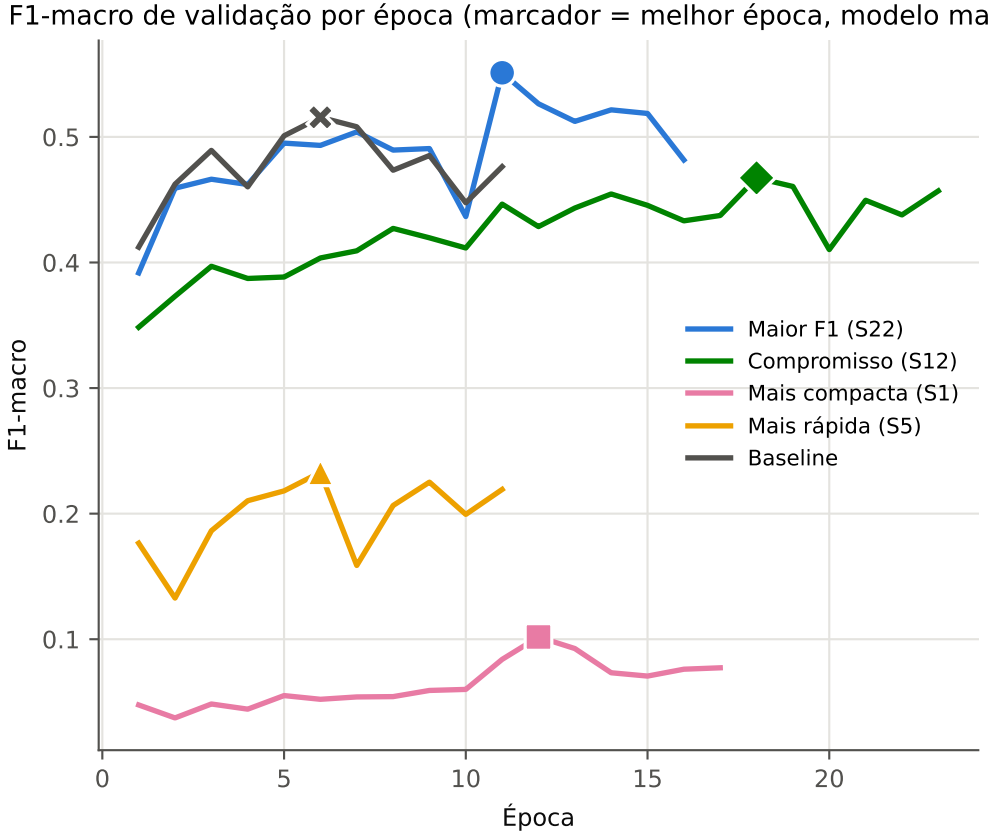

Desempenho por base na validação — soluções escolhidas antes do teste
Maior F1 (S22)
Compromisso (S12)
Mais compacta (S1)
Mais rápida (S5)
Baseline
Acurácia
F1-macro (classes presentes na base)
TNR — benigno classificado como benigno
TPR — ataque detectado como ataque
1.0
0.8
0.6
0.4
0.2
0.0
UNSW
ToN
CIC
Conjunto
PARCIAL | geração 71/100 | seed 42 | somente validação

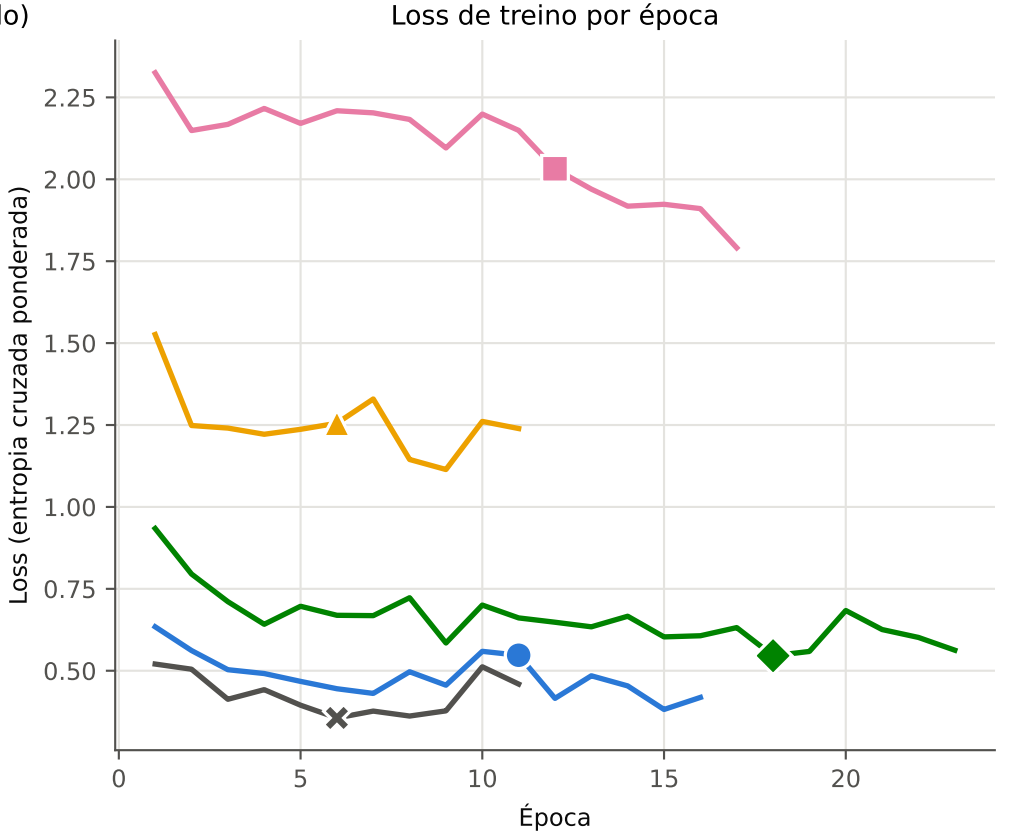
Classes na validação conjunta — prefixo indica a base com mais exemplos da classe

	Recall por classe (quanto da classe foi reconhecido)					F1 por classe (recall e precisão juntos)				
	Maior F1 (S22)	Compromisso (S12)	Mais compacta (S1)	Mais rápida (S5)	Baseline	Maior F1 (S22)	Compromisso (S12)	Mais compacta (S1)	Mais rápida (S5)	Baseline
todas · benign (n=2.837.545)	0.46	0.38	0.04	0.31	0.43	0.62	0.55	0.07	0.47	0.60
UNSW · exploits (n=4.567)	0.76	0.70	0.92	0.51	0.80	0.56	0.54	0.09	0.14	0.66
UNSW · fuzzers (n=2.880)	0.81	0.41	0.00	0.49	0.66	0.22	0.19	0.00	0.06	0.24
UNSW · generic (n=882)	0.60	0.49	0.00	0.00	0.61	0.09	0.03	0.00	0.00	0.03
UNSW · reconnaissance (n=774)	0.70	0.76	0.00	0.65	0.75	0.63	0.10	0.00	0.01	0.62
UNSW · shellcode (n=212)	0.59	0.40	0.00	0.06	0.72	0.01	0.03	0.00	0.07	0.01
UNSW · backdoor (n=181)	0.43	0.38	0.01	0.00	0.44	0.12	0.00	0.00	0.00	0.11
UNSW · analysis (n=113)	0.93	0.80	0.00	0.00	0.95	0.03	0.04	0.00	0.00	0.13
UNSW · worms (n=28)	0.18	0.21	0.00	0.00	0.18	0.06	0.10	0.00	0.00	0.02
ToN · xss (n=367.219)	0.91	0.77	0.00	0.25	0.92	0.87	0.69	0.00	0.31	0.86
ToN · ddos (n=231.053)	0.89	0.94	0.07	0.95	0.94	0.88	0.80	0.09	0.55	0.89
ToN · password (n=144.410)	0.77	0.81	0.00	0.54	0.87	0.80	0.73	0.00	0.56	0.83
ToN · dos (n=105.686)	0.91	0.85	0.00	0.67	0.87	0.79	0.61	0.00	0.22	0.80
ToN · injection (n=103.324)	0.76	0.49	0.84	0.00	0.78	0.63	0.52	0.28	0.00	0.73
ToN · scanning (n=87.417)	0.69	0.65	0.14	0.67	0.64	0.62	0.64	0.23	0.59	0.64
ToN · mitm (n=1.126)	0.38	0.38	0.91	0.38	0.44	0.06	0.01	0.00	0.00	0.01
ToN · ransomware (n=398)	0.97	0.78	0.05	0.55	0.82	0.05	0.08	0.00	0.01	0.04
CIC · ddos attack-hoic (n=157.070)	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.85	0.87	0.99
CIC · bot (n=132.631)	1.00	1.00	1.00	0.99	1.00	0.99	0.99	0.27	0.67	1.00
CIC · dos attacks-hulk (n=64.748)	1.00	1.00	1.00	0.98	1.00	1.00	0.99	0.63	0.91	1.00
CIC · ssh-bruteforce (n=21.471)	1.00	1.00	1.00	1.00	1.00	0.99	0.98	0.54	0.93	0.99
CIC · ddos attacks-loic-http (n=13.931)	1.00	0.99	0.82	0.82	1.00	0.96	0.96	0.09	0.09	0.96
CIC · infilteration (n=13.873)	0.83	0.65	0.04	0.16	0.81	0.02	0.02	0.01	0.02	0.02
CIC · dos attacks-slowhttptest (n=5.888)	1.00	1.00	0.00	1.00	1.00	1.00	0.99	0.00	0.73	0.99
CIC · dos attacks-goldeneye (n=4.056)	1.00	1.00	0.00	0.00	1.00	0.95	0.98	0.00	0.00	0.96
CIC · dos attacks-slowloris (n=2.349)	1.00	1.00	0.00	0.00	1.00	0.96	0.91	0.00	0.00	0.94
CIC · brute force -web (n=369)	0.22	0.21	0.00	0.00	0.21	0.16	0.04	0.00	0.00	0.03
CIC · ddos attack-loic-udp (n=299)	0.97	0.97	0.00	1.00	0.97	0.97	0.94	0.00	0.01	0.69
CIC · brute force -xss (n=162)	0.16	0.09	0.00	0.00	0.19	0.02	0.03	0.00	0.00	0.04
CIC · sql injection (n=61)	0.03	0.00	0.00	0.00	0.10	0.03	0.00	0.00	0.00	0.00
CIC · ftp-bruteforce (n=1)	1.00	1.00	0.00	0.00	1.00	1.00	0.01	0.00	0.00	0.17



PARCIAL | geração 71/100 | seed 42 | somente validação





PARCIAL | geração 71/100 | seed 42 | somente validação